AMERICAN UNIVERSITY OF BEIRUT

How Stable Are Regression Conclusions Across Subpopulations?

A Finite-Sample Empirical Study

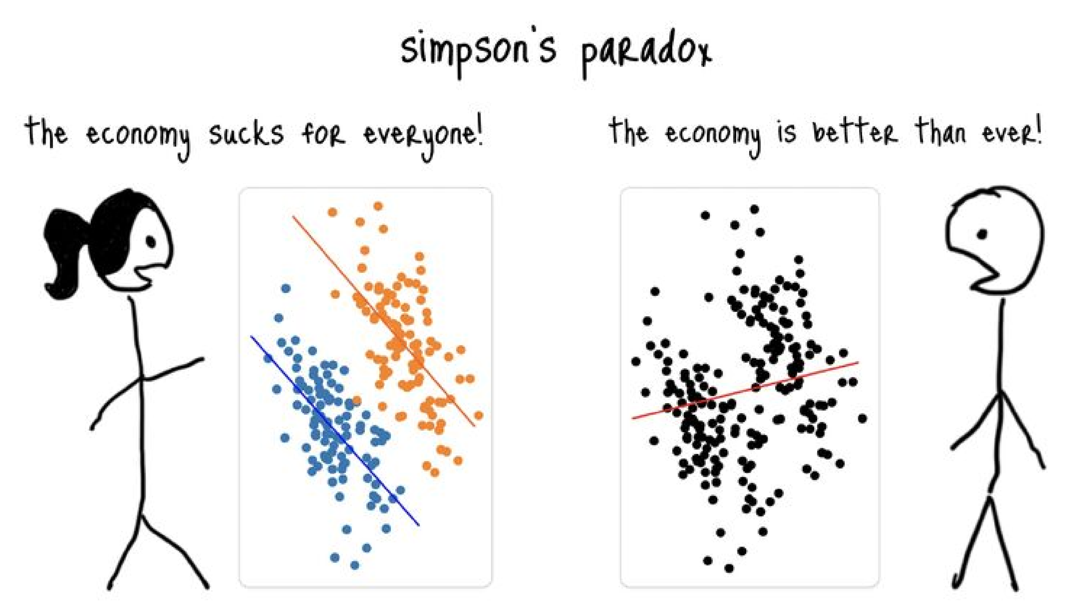
[Authors] – Department of Statistics – July 2026

The Problem

You fit a regression on the full dataset. R^2 is fine, p-values are small. A reviewer asks: *"Does this hold for each subgroup?"*

You check. It doesn't. Now what?

Simpson's Paradox



Within each group the trend is positive. Pooled, it reverses. The pooled regression hides subgroup structure.

"It's an Artifact"

"The coefficient
across grou

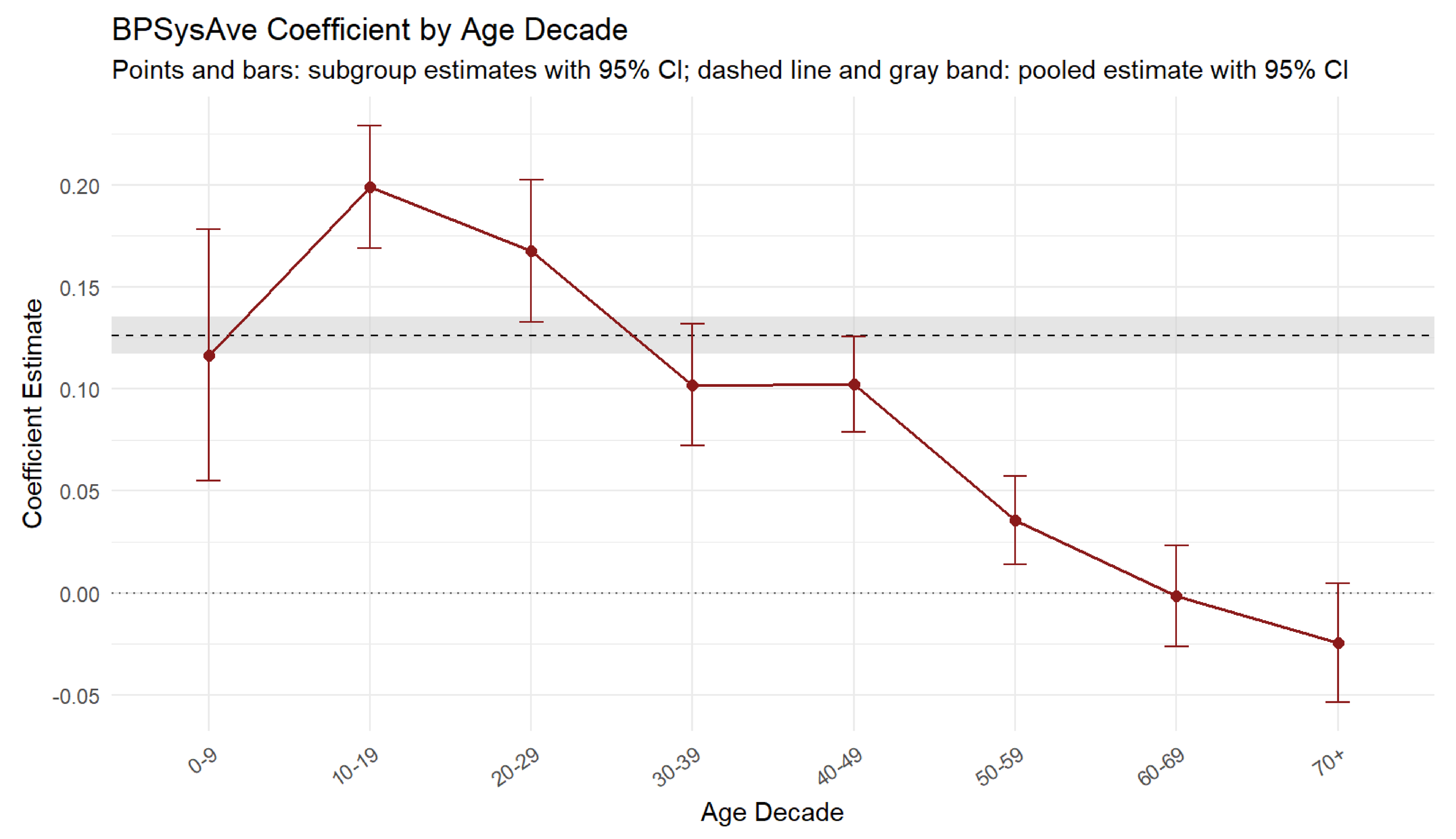
"Genuine hete
Model doesn't

"Wait — different X
distributions cause
most of the drift."

Coefficient differences come from **two sources**: genuine heterogeneity and mechanical artifacts from covariate distribution differences. Standard practice conflates them.

Real-Data Evidence: NHANES Blood Pressure

Data: NHANES 2017–2018 ($n = 9{,}254$), systolic BP response, age-decade subgroups. Predictors: BMI, waist circumference, total cholesterol, HDL.



Metric	Value
SSR (all predictors)	0.57
Max standardized drift	32.07 (BPSysAve)
Context-norm. drift	7.49 (BPSysAve)
Avg. CSS (pairwise)	0.28

Key finding: Waist circumference's effect on BP *reverses sign* in older groups. Drift 32→7.5 after context normalization, confirming genuine heterogeneity.

Five Stability Metrics

Metric	Definition	What it captures
SSR	$\mathrm{SSR}_j = \frac{1}{M}\sum \mathbf{1}(\mathrm{sgn}\,\hat\beta_{j,a} = \mathrm{sgn}\,\hat\beta_{j,b})$	Do conclusions $(+/-)$ reverse across groups?
Standardized drift	$D^\star_j = \frac{\max_g \lvert\hat\beta_{j,g}-\hat\beta_{j,\mathrm{pooled}}\rvert}{\widehat{\mathrm{SE}}(\hat\beta_{j,\mathrm{pooled}})}$	How far do subgroup coefficients move from pooled, in SE units?
Context-norm. drift	$\widetilde{D}_j = \dfrac{D^\star_j}{1+M_j}$, $M_j = \frac{1}{\binom{G}{2}}\sum_{a<b}\frac{\lvert\bar x_{j,a}-\bar x_{j,b}\rvert}{s_{j,\mathrm{pooled}}}$	Drift after accounting for covariate distribution differences.
TPG	$\mathrm{TPG}_{a\to b} = \mathrm{RMSE}_{\mathrm{pooled}\to b} - \mathrm{RMSE}_{b\to b}$	Prediction loss when a model travels across groups.
CSS	$\mathrm{CSS}_{a,b} = \frac{1}{p}\sum_j \frac{\lvert\bar x_{j,a}-\bar x_{j,b}\rvert}{s_{j,\mathrm{pooled}}}$	How different are X distributions between groups?

Key Decomposition: Why Context Drift < Raw Drift

Even with identical β, different X distributions inflate drift:

$$\hat\beta^{(g)} - \hat\beta^{(\mathrm{pooled})} = (X_g'X_g)^{-1}X_g'\varepsilon_g - (X'X)^{-1}X'\varepsilon$$

$$\underbrace{D^\star_j}_{\text{Total}} = \underbrace{\widetilde{D}_j}_{\text{Genuine}} \times \underbrace{(1+M_j)}_{\text{Mechanical}}$$

3 regimes: $M_j \approx 0 \Rightarrow$ all drift real; large M_j + small $\widetilde{D}_j \Rightarrow$ artifact; both large $\Rightarrow$ genuine heterogeneity.

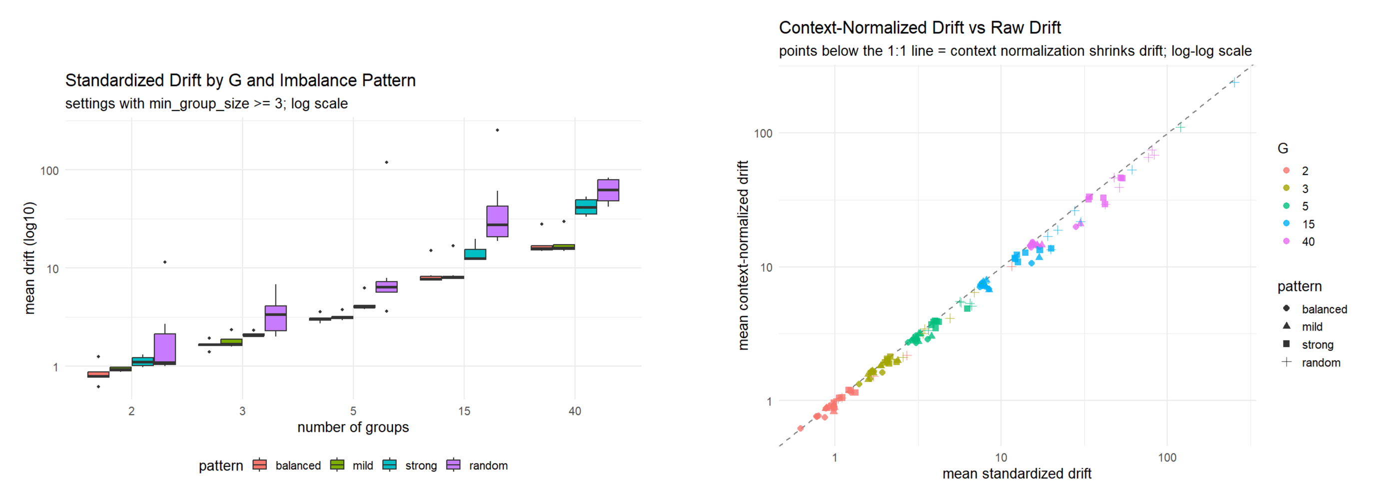
Four Models

Model	Specification
M1 (Pooled)	$Y = \beta_0 + X^\top\beta + \varepsilon$
M2 (Intercept)	$+\gamma G$
M3 (Interaction)	$+\gamma G + X^\top\delta G$
M4 (Separate)	Coeffs per group

Isolates: intercept shift → slope heterogeneity → full subgroup structure.

Simulation Grid (Scenario 1)

140 settings: $G \in \{3, 5, 10, 20, 40\}$, $n \in \{100, \ldots, 30000\}$, 4 imbalance patterns (balanced, mild, strong, Dirichlet-random). 200 reps each.



Drift grows with G and imbalance (left). Context normalization shrinks drift (right).

What Drives Instability?

- Drift driven by G and per-group n, not total N.
- Strong/random imbalance amplifies drift 2–4× over balanced.
- SSR = 1.0 throughout (Scenario 1: identical β, no sign flips).
- Context normalization reduces drift; shrinkage ∝ CSS.

Future Work

Same analysis for Scenarios 2–4, then:

1. Introduce **collinearity** across predictors
2. Scale to **higher dimensions** (more predictors)
3. Test **different regression models** beyond OLS

References

Berk et al. (2019); Tsoi et al. (2026); Bertsimas & Paskov (2020); Liu et al. (2020); Watson & Engle (1985); Garbade (1977); Buja et al. (2006).

`github.com/Nadjoubiii/Regression-conclusion-stability-paper`